

**Tool: R Programming**

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**Subject: Data Handling & Visualization**

**Code: DSA0604**

## Scenario 1: Student Academic Performance Dashboard

### Code:

```
library(ggplot2)

data <- data.frame(

  Attendance = c(85,90,75,60,95,80),

  Marks = c(80,90,70,55,95,78),

  Department = c("CSE","AI","IT","ECE","CSE","AI")

)

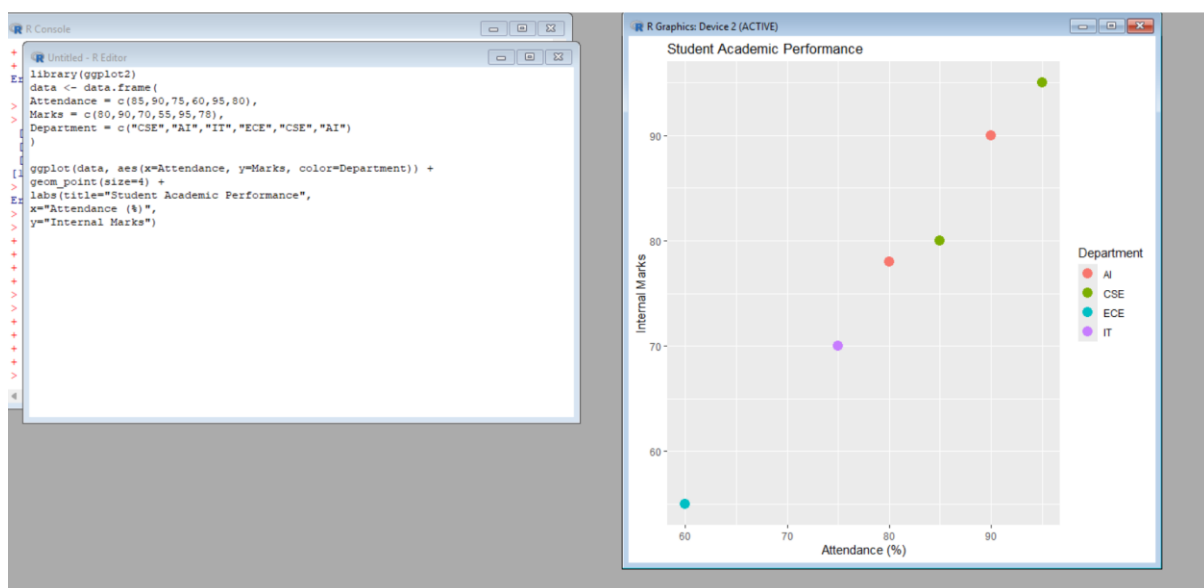
ggplot(data, aes(x=Attendance, y=Marks, color=Department)) +

  geom_point(size=4) +

  labs(title="Student Academic Performance",

        x="Attendance (%)",

        y="Internal Marks")
```



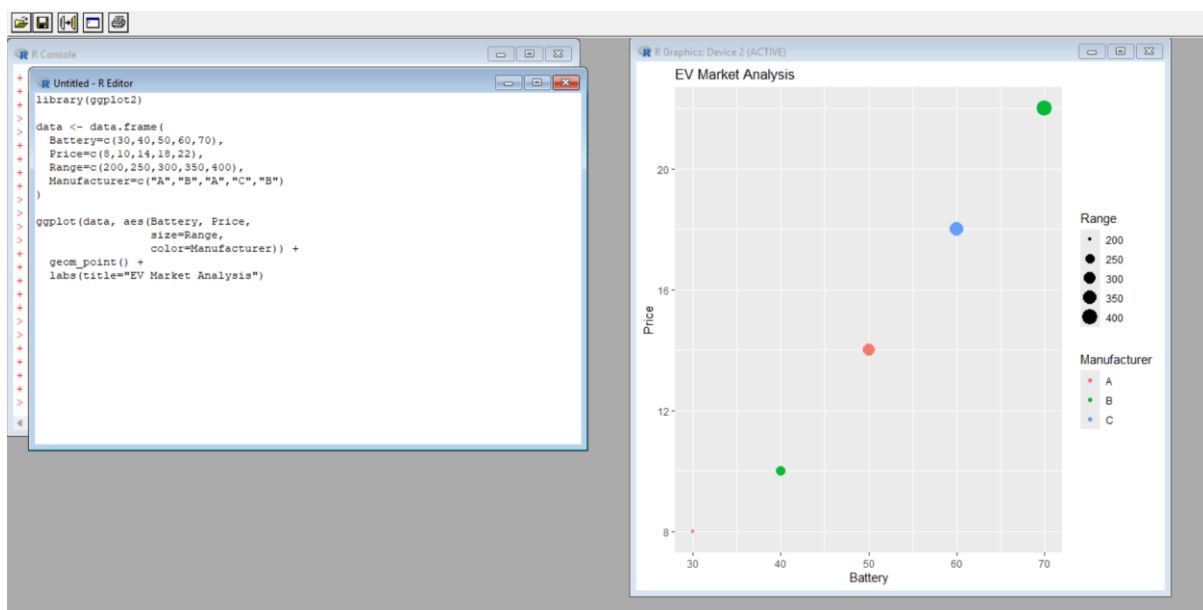
## Scenario 2: Electric Vehicle Market Analysis

### Code:

```
library(ggplot2)

data <- data.frame(
  Battery=c(30,40,50,60,70),
  Price=c(8,10,14,18,22),
  Range=c(200,250,300,350,400),
  Manufacturer=c("A","B","A","C","B")
)

ggplot(data, aes(Battery, Price,
  size=Range,
  color=Manufacturer)) +
  geom_point() +
  labs(title="EV Market Analysis")
```



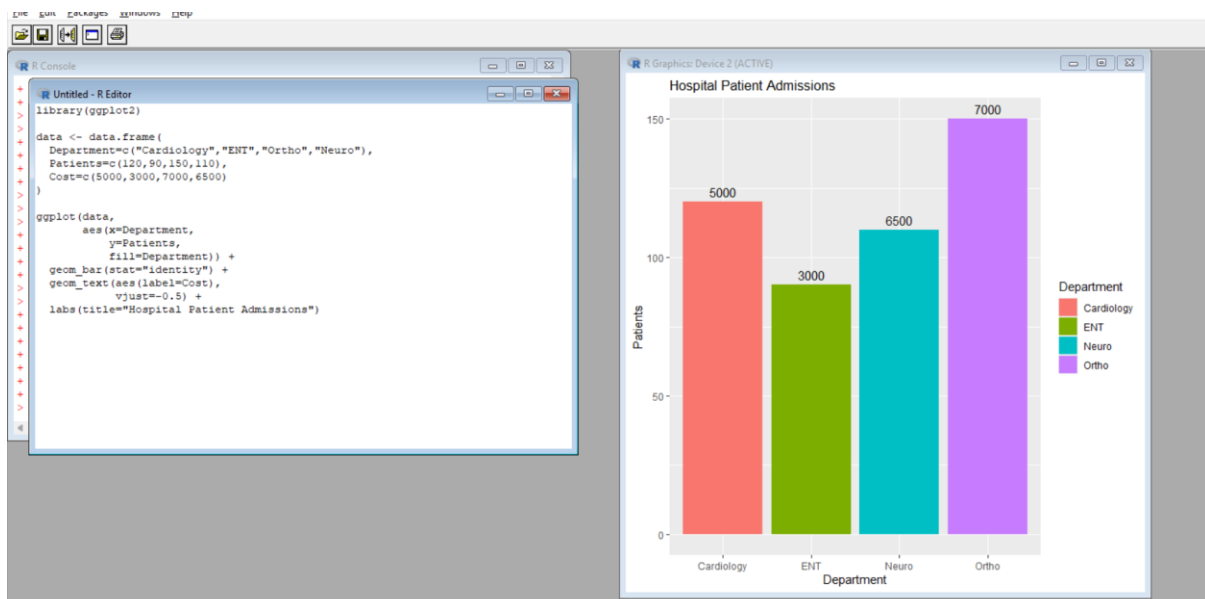
### Scenario 3: Hospital Patient Admission Analysis

#### Code:

```
library(ggplot2)

data <- data.frame(
  Department=c("Cardiology","ENT","Ortho","Neuro"),
  Patients=c(120,90,150,110),
  Cost=c(5000,3000,7000,6500)
)

ggplot(data,
  aes(x=Department,
      y=Patients,
      fill=Department)) +
  geom_bar(stat="identity") +
  geom_text(aes(label=Cost),
    vjust=-0.5) +
  labs(title="Hospital Patient Admissions")
```



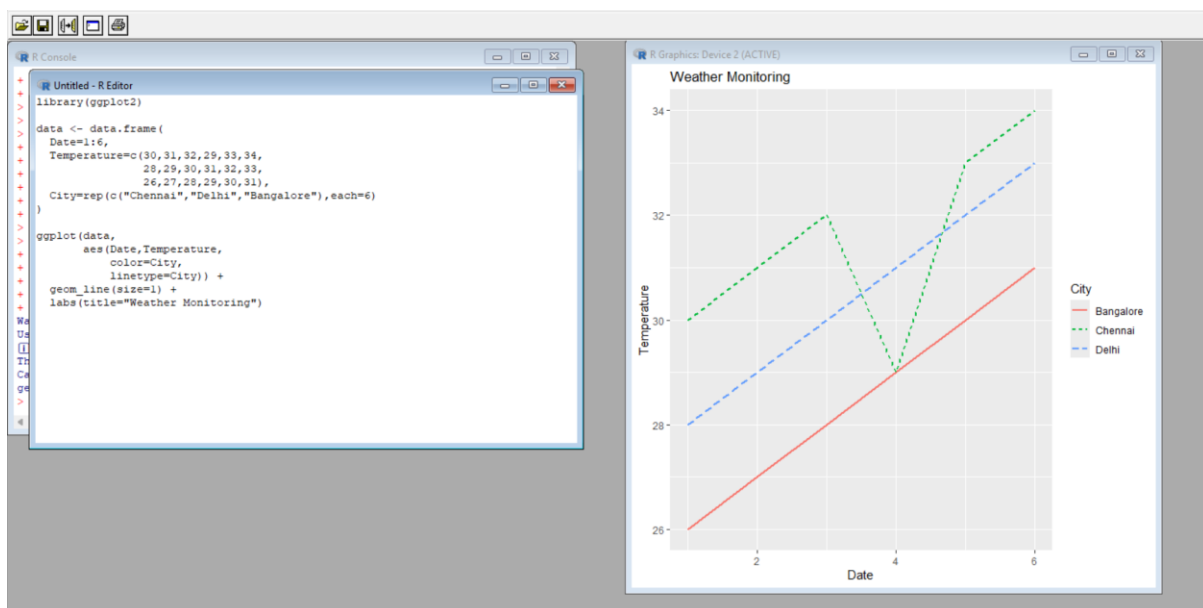
## Scenario 4: Weather Monitoring System

### Code:

```
library(ggplot2)

data <- data.frame(
  Date=1:6,
  Temperature=c(30,31,32,29,33,34,
                28,29,30,31,32,33,
                26,27,28,29,30,31),
  City=rep(c("Chennai","Delhi","Bangalore"),each=6)
)

ggplot(data,
  aes(Date, Temperature,
        color=City,
        linetype=City)) +
  geom_line(size=1) +
  labs(title="Weather Monitoring")
```



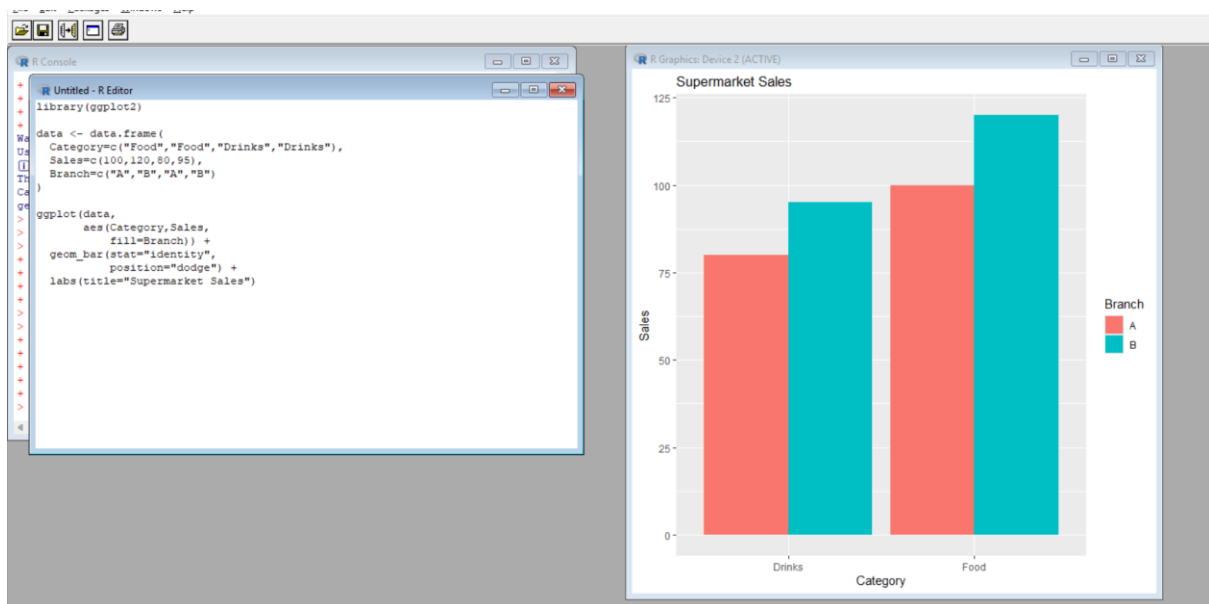
## Scenario 5: Supermarket Sales Analysis

### Code:

```
library(ggplot2)

data <- data.frame(
  Category=c("Food","Food","Drinks","Drinks"),
  Sales=c(100,120,80,95),
  Branch=c("A","B","A","B")
)

ggplot(data,
  aes(Category,Sales,
    fill=Branch)) +
  geom_bar(stat="identity",
    position="dodge") +
  labs(title="Supermarket Sales")
```



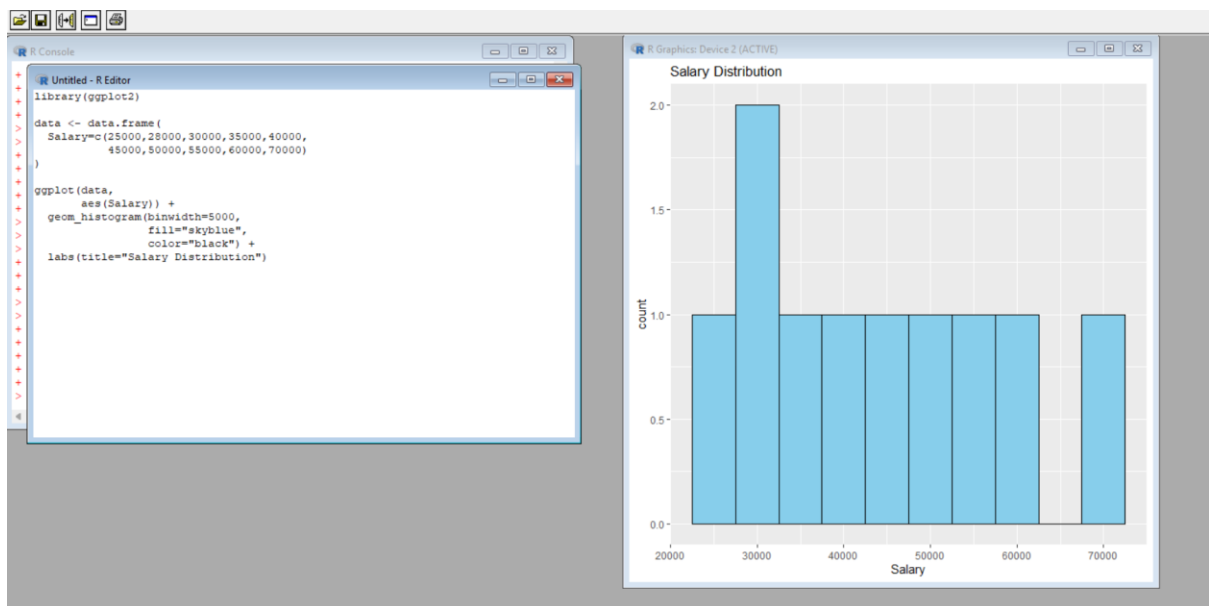
## Scenario 6: Employee Salary Distribution

### Code:

```
library(ggplot2)

data <- data.frame(
  Salary=c(25000,28000,30000,35000,40000,
           45000,50000,55000,60000,70000)
)

ggplot(data,
  aes(Salary)) +
  geom_histogram(binwidth=5000,
    fill="skyblue",
    color="black") +
  labs(title="Salary Distribution")
```



## Scenario 7: Smartphone Market Comparison

### Code:

```
library(ggplot2)

data <- data.frame(

  RAM=c(4,6,8,12,16),

  Price=c(15000,22000,28000,45000,60000),

  Brand=c("Samsung","Vivo","Realme","Apple","OnePlus"),

  Storage=c(64,128,128,256,512),

  Speed=c(2.0,2.2,2.5,3.0,3.2)

)

ggplot(data,

  aes(RAM,Price,

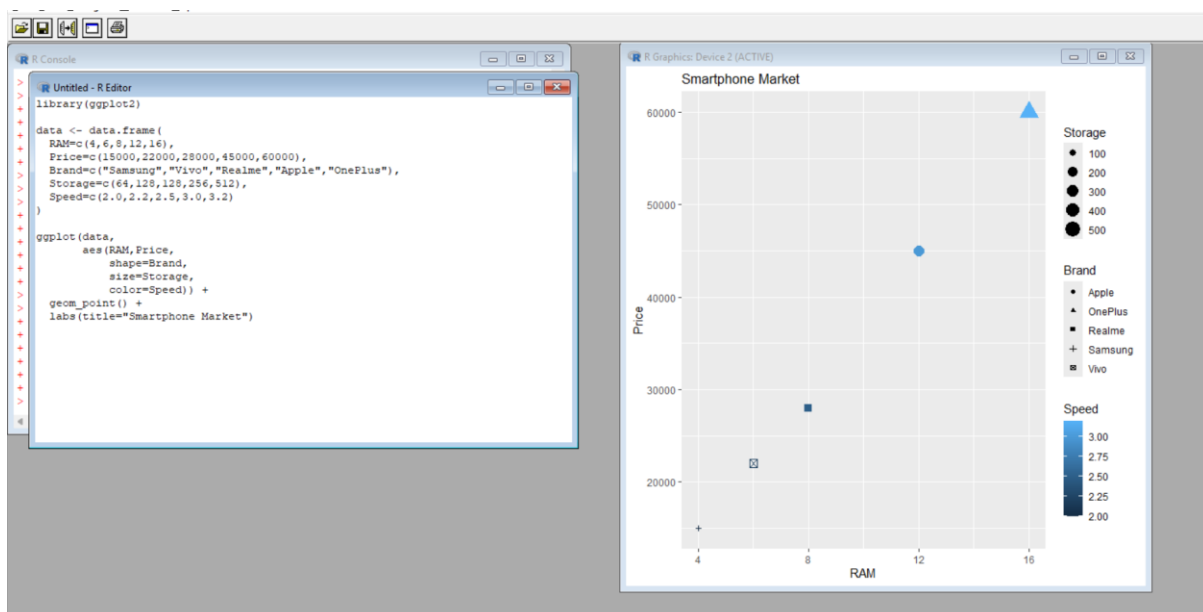
    shape=Brand,

    size=Storage,

    color=Speed)) +

  geom_point() +

  labs(title="Smartphone Market")
```



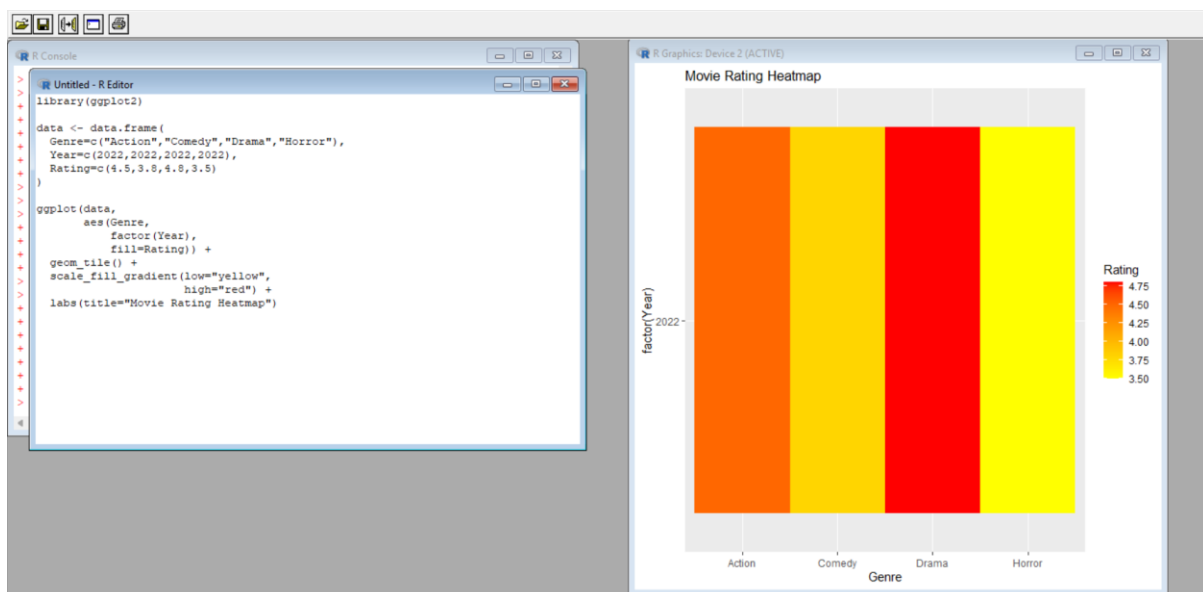
## Scenario 8: Online Movie Rating Analysis

### Code:

```
library(ggplot2)

data <- data.frame(
  Genre=c("Action","Comedy","Drama","Horror"),
  Year=c(2022,2022,2022,2022),
  Rating=c(4.5,3.8,4.8,3.5)
)

ggplot(data,
  aes(Genre,
      factor(Year),
      fill=Rating)) +
  geom_tile() +
  scale_fill_gradient(low="yellow",
                      high="red") +
  labs(title="Movie Rating Heatmap")
```





## Scenario 9: COVID-19 State-wise Analysis

### Code:

```
library(ggplot2)

data <- data.frame(

  State=c("Tamil Nadu","Kerala","Karnataka","Andhra"),

  Cases=c(2000,3500,2800,1500)

)

ggplot(data,

  aes(State,"COVID",

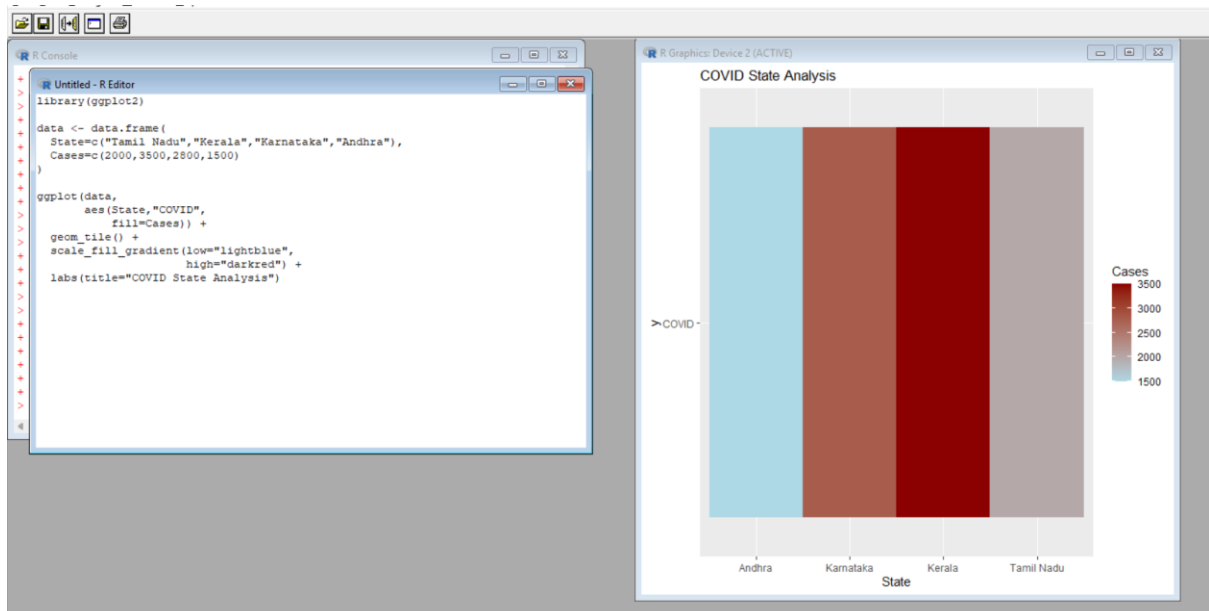
    fill=Cases)) +

  geom_tile() +

  scale_fill_gradient(low="lightblue",

    high="darkred") +

  labs(title="COVID State Analysis")
```



## Scenario 10: Iris Flower Classification

### Code:

```
library(ggplot2)

data(iris)

ggplot(iris,
       aes(Sepal.Length,
           Petal.Length,
           color=Species,
           shape=Species,
           size=Sepal.Width)) +
  geom_point() +
  labs(title="Iris Flower Classification")
```

